



TOURO
UNIVERSITY

Imaging and Clinical Context

Bone/ Skeletal System



"To succeed in life, you need three things: a wishbone, a backbone and a funny bone."

-Reba McEntire



Imaging Modalities Quick Review

X-Ray Imaging

REVIEW: MECHANISM OF ACTION/ PROS & CONS

Photon Beam: X-ray machine emits a controlled beam of X-ray photons, which pass through the body onto a detector on the other side

Image Creation: Detector captures the X-rays that have traversed the body and converts them into an electrical signal which is then processed into a digital image

Pros	Cons
Versatility: Diagnosing fractures, detecting infections, and guiding medical procedures Speed of Imaging Widespread Availability Cost-Effectiveness	Limited Soft Tissue Detail Contrast Sensitivity: X-ray images may not always provide sufficient contrast between different tissues or structures Not Suitable for All Conditions: Soft tissue injuries or diagnosing certain types of tumors Risk to Pregnant Women

Ultrasound Imaging

REVIEW: MECHANISM OF ACTION/ PROS & CONS

- Transducer:** Emits high-frequency sound waves
- Reflection and Absorption:** Tissues with **higher density**, like **bone**, reflect more sound waves, while less dense tissues, like fluid-filled organs, allow more sound waves to pass through
- Echo Detection:** The transducer receives the echoes produced by the sound waves bouncing off the internal structures.
- Image Formation:** The echoes detected are processed by a computer to generate real-time images

Pros	Cons
Non-invasive Real-time imaging No radiation exposure Portability Cost-effective Versatility: Visualize soft tissues, muscles, tendons, blood vessels, heart, liver, kidneys, and reproductive organs	Limited penetration and resolution Operator-dependent Limited tissue characterization Challenges with obesity and air-filled structures Difficulty in imaging certain areas: CNS Limited availability of 3D/ volumetric imaging

Computed Tomography (CT) Imaging

REVIEW: MECHANISM OF ACTION/ PROS & CONS

X-ray Source and Detector Array: CT imaging involves a rotating X-ray source and a ring-shaped detector array that surrounds the patient

Data Acquisition: Multiple X-ray projections collected

Data Reconstruction: Mathematical algorithms are applied to the raw data to reconstruct cross-sectional images (slices) of the body.

Image Display

Pros	Cons
Detailed Visualization Speed Versatility: Image the brain, chest, abdomen, pelvis, musculoskeletal system, tumors, fractures, and evaluate for internal bleeding Non-invasive Availability Emergency Use: Rapid diagnosis of traumatic injuries, strokes, and internal bleeding	Radiation Exposure Contrast Material Risks Cost: More expensive than X-rays or ultrasound Limited Soft Tissue Contrast: Compared to magnetic resonance imaging (MRI) Artifact Formation: Patient motion or metallic implants Not Suitable for Pregnant Patients Overutilization

ORIENTATION OF IMAGING

It is important to note that current CT machines display the image opposite the patient's side as the image is produced as viewed from the patient's foot. Thus the right side of the image is the patient's left side.



Magnetic Resonance (MRI) Imaging

REVIEW: MECHANISM OF ACTION/ PROS & CONS

Magnetic Field Alignment: Produced by a superconducting magnet which aligns the hydrogen atoms (protons) in the body along its direction.

Radiofrequency (RF) Excitation: Radiofrequency (RF) energy is then applied to the body, causing the aligned hydrogen atoms to absorb energy and temporarily deviate from their aligned state

Relaxation: Hydrogen atoms gradually return to their original aligned state and emit radiofrequency signals as they release the absorbed energy.

Signal Detection: The emitted radiofrequency signals are detected by receiver coils providing information about the density & distribution of hydrogen atoms in different tissues.

Spatial Encoding: MRI scanner can encode spatial information into the signals received from the patient's body.

Image Reconstruction: The signals detected by the receiver coils are processed by a computer to reconstruct cross-sectional images of the body into detailed images.

Image Display

Magnetic Resonance (MRI) Imaging

REVIEW: MECHANISM OF ACTION/ PROS & CONS

Pros	Cons
Non-invasive High-resolution images Versatility: Strong images of brain, spine, joints, and organs Multi-planar imaging: (sagittal, coronal, and axial) Functional Imaging Available: Measures changes in blood flow in the brain, allowing study of brain function and activity in real-time	Cost Time-consuming: Take longer to perform compared to other imaging modalities Contraindications: Imaging can interfere with certain medical devices (such as pacemakers or metal implants) and may not be safe for pregnant women (especially during the first trimester) Claustrophobia Limited availability

Course Objectives



- Describe the basic anatomy of bone—including cortex, medulla, endosteum, and the key regions (epiphysis, metaphysis, diaphysis apophysis).
- Differentiate imaging modalities (X-ray, CT, MRI, ultrasound) and explain their mechanisms, strengths, and limitations in skeletal evaluation.
- Systematically describe fractures using the five key descriptors and recognize common fracture patterns (transverse, spiral, comminuted, segmental, intra-articular, impacted, compression, stress).
- Identify pediatric fracture principles, including Salter-Harris classification and the biomechanical differences between pediatric and adult bone.

Course Objectives



- Recognize and interpret pathologic fractures, differentiate bone cancers from metastatic lesions based on patient age, type, and imaging characteristics.
- Explain the biological process of bone healing and the factors that enhance or delay repair, correlating radiologic changes with physiologic stages.
- Apply clinical reasoning by linking fracture location to potential neurovascular injury and formulating appropriate next diagnostic or management steps.

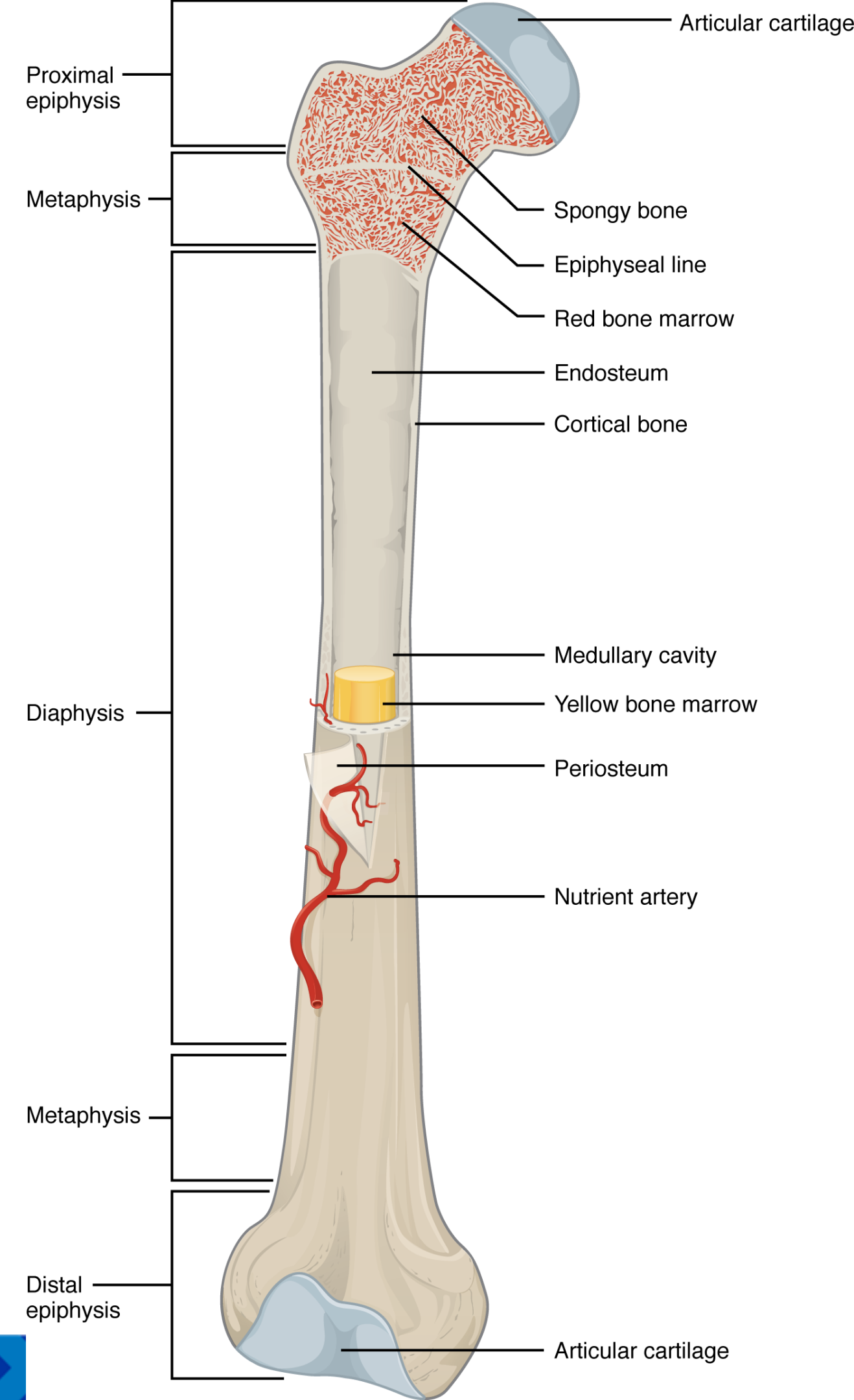
Review of Bone Anatomy

Cortex (Cortical/Compact Bone):

- Dense, strong outer shell.
- Provides mechanical strength and structural support.
- Appears white and solid on radiographs.
- Found mainly in diaphyses of long bones.

Medulla (Cancellous/Trabecular/Spongy Bone):

- Located inside the cortex.
- Contains a lattice-like trabecular network filled with bone marrow.
- Important for shock absorption and metabolic activity
- More prominent in epiphyses and metaphyses.



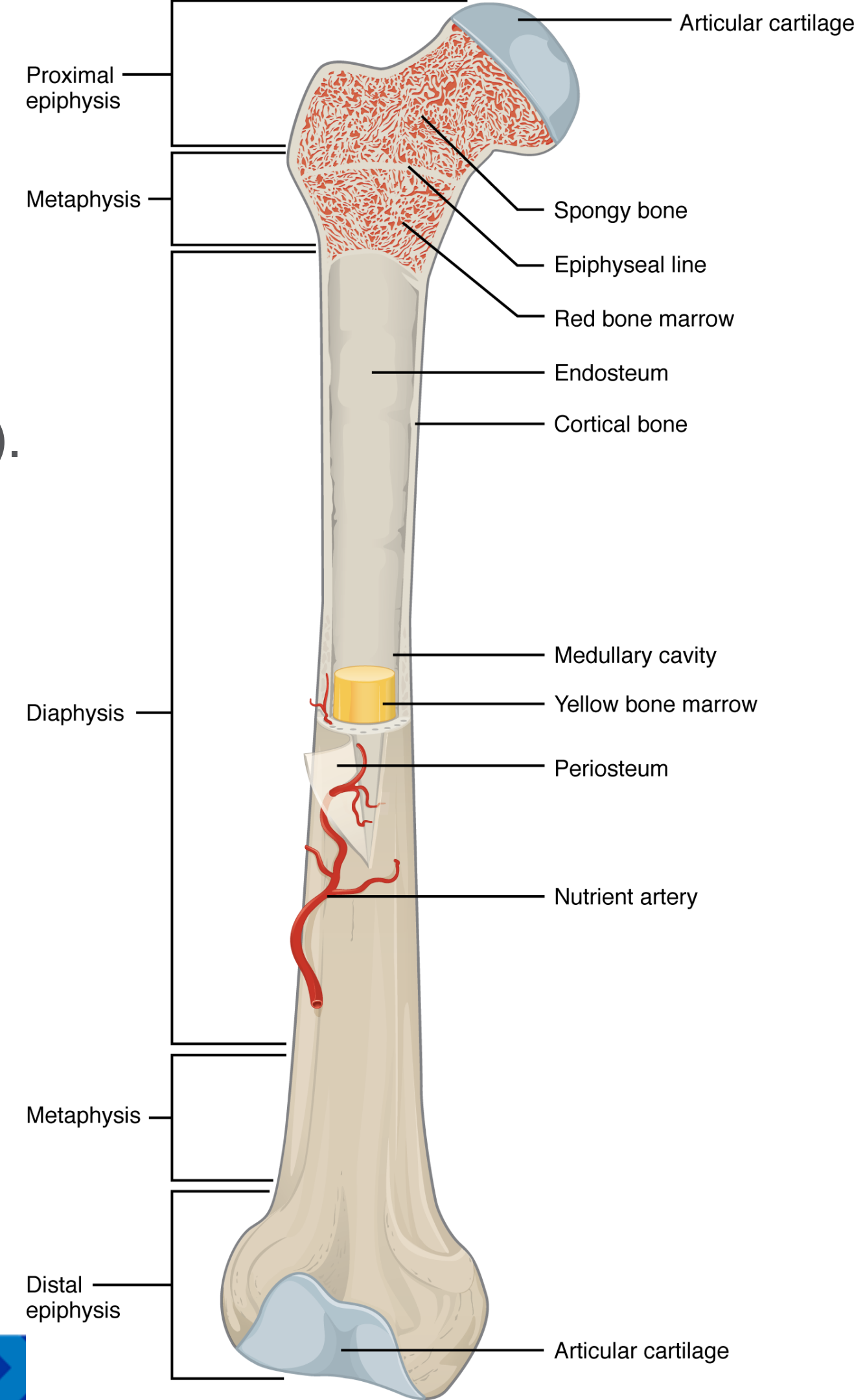
Review of Bone Anatomy

Periosteum:

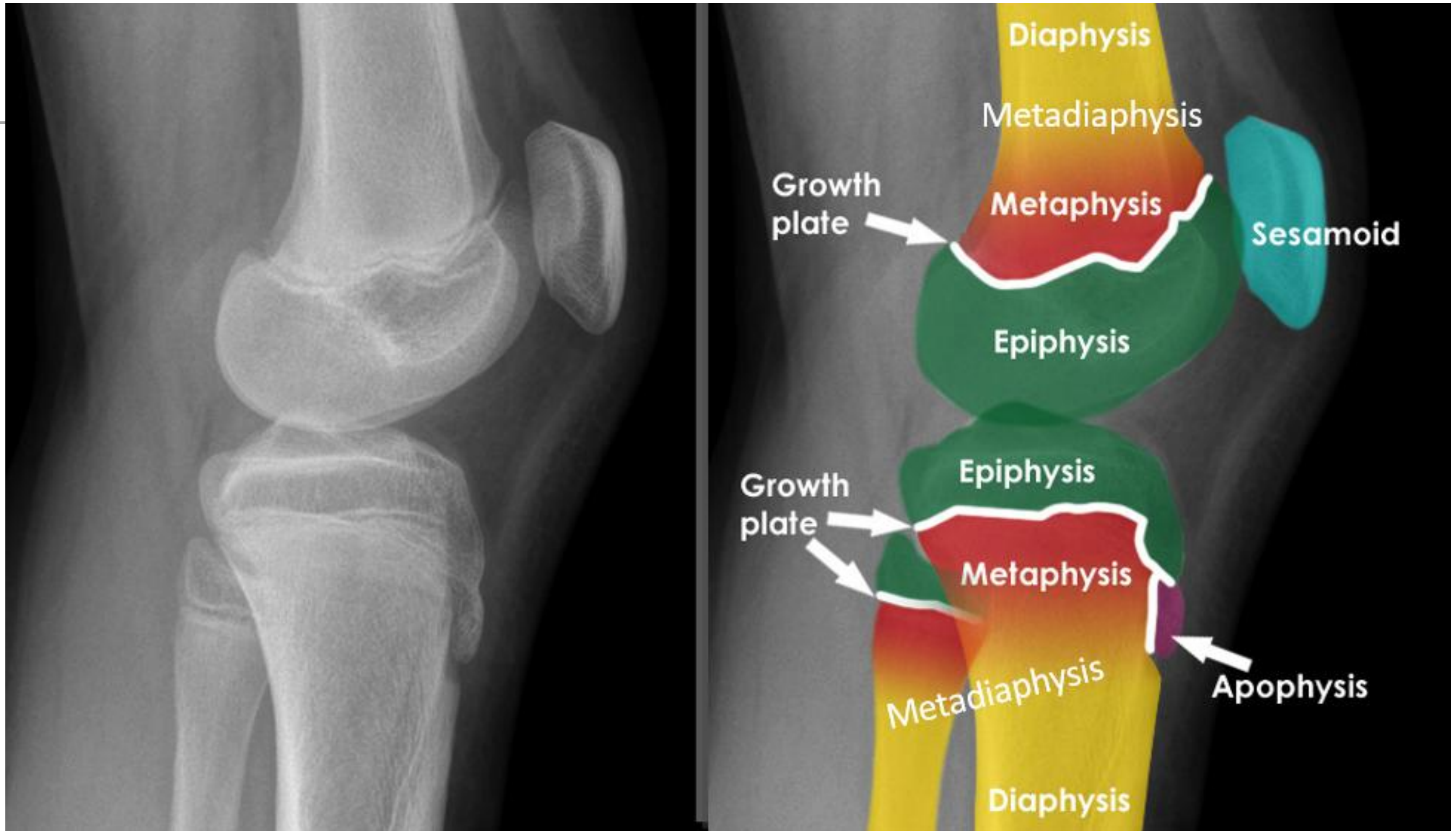
- Outer fibrous layer covering bone (except at articular surfaces).
- Rich in blood vessels, lymphatics, and nerves.
- Critical for bone growth (appositional) and healing after fractures.

Endosteum:

- Thin, vascular connective tissue lining the inner surface of bone (medullary cavity).
- Contains osteoblasts and osteoclasts → important for remodeling.



Key Anatomic Regions of Long Bones



Key Anatomic Regions of Long Bones

Epiphysis

- The rounded end of a long bone, located beyond the growth plate (physis).
- Primarily composed of **trabecular (spongy) bone** with a thin cortical shell.
- Functions in **joint articulation** and **load distribution**.

Metaphysis

- The flared portion of bone between the **diaphysis (shaft)** and the **epiphysis**.
- Rich in **trabecular bone** and highly vascularized.
- Common site for **growth (in children)** and **pathology** (e.g., osteomyelitis, bone tumors).

Key Anatomic Regions of Long Bones

Diaphysis

- The **shaft** of a long bone.
- Composed mainly of **cortical (compact) bone**, surrounding the medullary (marrow) cavity.
- Provides **structural strength** and acts as a lever arm for muscles.

Apophysis

- A **bony outgrowth** that serves as a tendon or ligament attachment site.
- Has its own ossification center but **does not contribute to longitudinal growth** (unlike the epiphysis).

Clinical Pearl

When ordering X-ray studies for evaluation of bones or joints:

Always obtain multiple views

Use at least two orthogonal (90°) projections to avoid missing fractures, dislocations, or foreign bodies.

- *Examples:*

- **Ankle:** AP, lateral, and mortise views
- **Knee:** AP and lateral ± sunrise (patellar) view
- **Shoulder:** AP and axillary or scapular Y views

Clinical Pearl

When ordering X-ray studies for evaluation of bones or joints:

Include the joint above and below the injury site when appropriate

This ensures detection of associated or referred injuries that may not be localized to a single region.

- **Forearm injury:** Order *“Forearm X-ray—AP and lateral, include wrist and elbow joints.”*
- **Tibia/fibula injury:** Order *“Tibia-fibula X-ray—AP and lateral, include knee and ankle joints.”*
- **Humeral shaft fracture:** Order *“Humerus X-ray—AP and lateral, include shoulder and elbow joints.”*

How to Describe a Fracture: Key Descriptors

When presenting a fracture, the description should be systematic, precise, and reproducible. Radiologists and orthopedists often use the following categories:

- Site
- Displacement
- Angulation
- Comminution
- Intra-/ Extra-Articular Involvement

How to Describe a Fracture: Key Descriptors

1. Site (Location)

- Which bone is involved (e.g., femur, radius, humerus).
- Which region of the bone:
 - Epiphysis (end)
 - Metaphysis (flared region)
 - Diaphysis (shaft)
- Example: “Mid-diaphyseal fracture of the humerus.”

How to Describe a Fracture: Key Descriptors

2. Displacement

- How far the fracture fragments have shifted from their normal alignment.
- Typically described in percent translation or centimeters.
 - Example: “Distal fragment displaced one shaft-width posteriorly.”
- If minimal: “Minimally displaced fracture.”

How to Describe a Fracture: Key Descriptors

3. Angulation

- Refers to the direction the distal fragment points relative to the proximal bone.
- Described in degrees and direction: apex medial, apex lateral, apex anterior, etc.
- Example: “20° apex-volar angulation of the distal radius.”

How to Describe a Fracture: Key Descriptors

4. Comminution

- Whether the bone is broken into multiple pieces.
- Types:
 - Simple (two fragments)
 - Comminuted (three or more fragments)
 - Segmental (two separate fractures creating a floating segment)
- Example: “Comminuted mid-shaft tibial fracture.”

How to Describe a Fracture: Key Descriptors

5. Intra-articular vs. Extra-articular

- Does the fracture extend into the joint surface?
- Intra-articular fractures are more complex and have higher risk of post-traumatic arthritis.
- Example:
 - Extra-articular: “Metaphyseal fracture of the distal radius without joint extension.”
 - Intra-articular: “Distal tibial plafond fracture with joint involvement.”

Let's Practice...

What do you see?



Site (Location): Mid-diaphysis of the left femur.

Fracture Pattern: Predominantly transverse with an oblique extension.

Displacement: Distal fragment displaced medially (nearly one shaft-width).

Angulation: Apex lateral (varus) angulation is present.

Comminution: Mild comminution is seen at the fracture site, with several small cortical fragments adjacent to the fracture.

Intra-/Extra-articular: The fracture is extra-articular, not extending into the hip or knee joint.



Putting it all together:

“Extra-articular, left femoral mid-diaphyseal fracture, transverse with oblique extension, displaced medially by approximately one shaft-width, with apex lateral angulation and mild comminution.”



Key Differences Between Pediatric and Adult Bone

Presence of Pysis (Growth Plate)

- Pediatric bone has a cartilaginous pysis responsible for longitudinal growth.
- This zone is structurally weaker and prone to injury → hence the need for *Salter-Harris classification*.

Periosteum Thickness & Healing Potential

- Pediatric periosteum is thick, strong, and highly osteogenic.
- Provides stability and rapid healing → fracture remodeling is much greater in children.

Key Differences Between Pediatric and Adult Bone

Bone Composition

- Pediatric bone has more collagen and water, less mineralization.
- More flexible and can absorb force, which explains incomplete fractures (greenstick).

Remodeling Potential

- Pediatric bones can correct angulation over time due to growth and remodeling.
- Adult bones remodel minimally and thus, malunion in adults is much less forgiving.

Why the Salter-Harris Classification Exists for Pediatrics (Not Adults)

The **Salter-Harris system** is used to classify fractures involving the **physis (growth plate)**, a structure **only present in children and adolescents**.

Adults lack a physis because once skeletal maturity is reached, the growth plate closes and is replaced by solid bone (epiphyseal line).

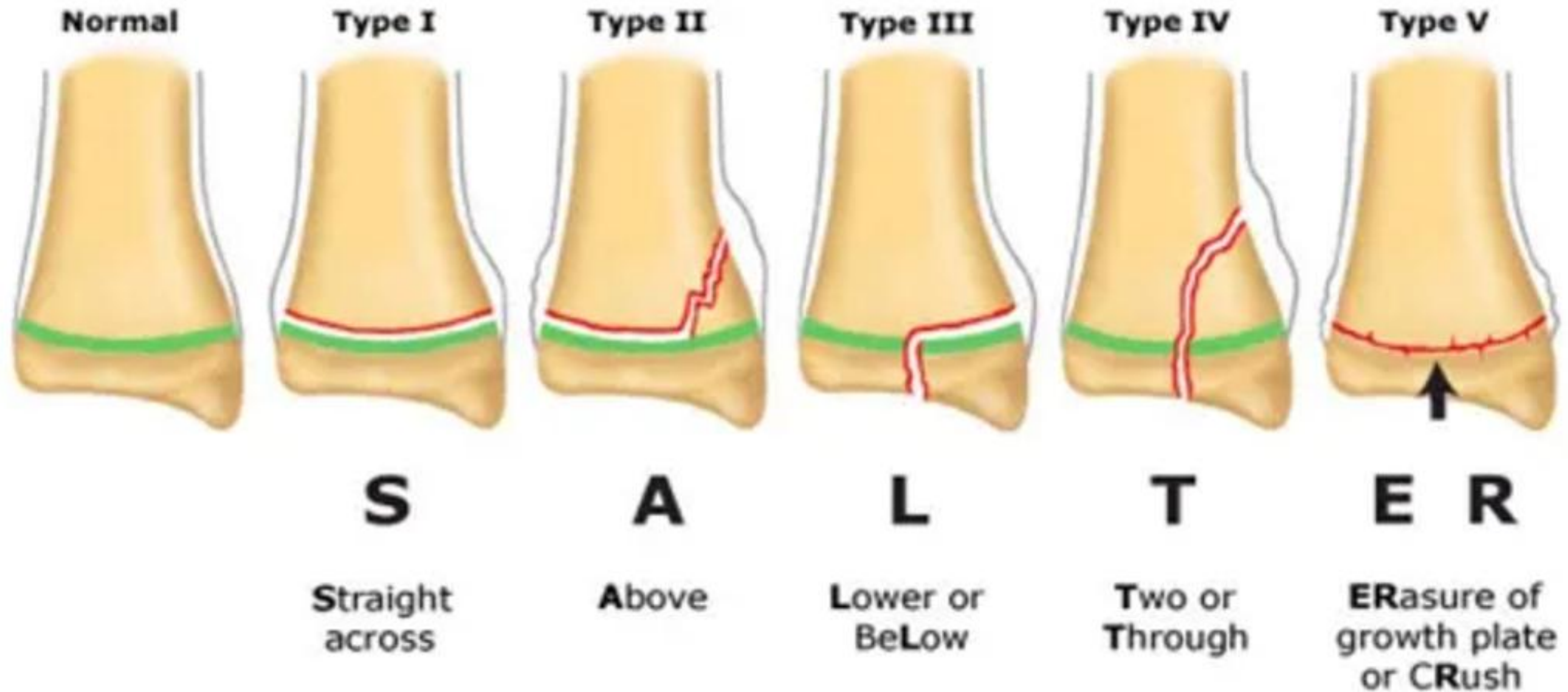
Salter-Harris Classification: The Basics

- **Type I:** Straight through the physis.
- **Type II:** Through the physis and metaphysis (most common).
- **Type III:** Through the physis and epiphysis (involving joint surface).
- **Type IV:** Through metaphysis, physis, and epiphysis.
- **Type V:** Crush injury to the physis (worst prognosis).

Mnemonic: **SALTER**

- **S**traight across (I)
- **A**bove (II – into metaphysis)
- **L**ower (III – into epiphysis)
- **T**hrough (IV – through metaphysis, physis, epiphysis)
- **E**Rasure/crush (V)

Salter-Harris classification of physeal fractures



Not all bone fractures are due to trauma...

Pathologic Fractures

A pathologic fracture is a break in a bone that occurs through an area weakened by disease, rather than by normal mechanical trauma.

Fractures can occur with minimal or no trauma

Atraumatic fractures or fractures not compatible with the mechanism of injury should always raise suspicion for underlying pathology: *metabolic bone disease, infection, or malignancy.*

Causes of Pathologic Fractures

Metabolic Bone Disease

- Osteoporosis
- Osteomalacia/ Rickets
- Paget's disease
- Hyperparathyroidism

Infection

- Osteomyelitis (especially chronic, with cortical destruction)

Neoplastic Disease (Malignancy or Benign Tumors)

- Primary bone cancers
- Metastatic lesions
- Benign tumors (e.g., unicameral bone cysts, giant cell tumor, fibrous dysplasia)

Primary Bone Cancers

Relatively rare compared to metastases.

Tend to occur in **younger patients** (teens, young adults).

Examples:

- **Osteosarcoma:** Metaphyseal regions of long bones, often around the knee; aggressive periosteal reaction (“sunburst,” Codman triangle).
- **Ewing sarcoma:** Diaphyseal lesions, onion-skin periosteal reaction; systemic symptoms (fever, pain).
- **Chondrosarcoma:** Typically in adults; cartilage-producing tumor.

Fractures occur due to tumor invasion weakening the cortical structure.

Metastatic Bone Lesions

Most common cause of pathologic fractures in adults.

Cancers that metastasize to bone:

- Breast
- Prostate
- Lung
- Kidney
- Thyroid

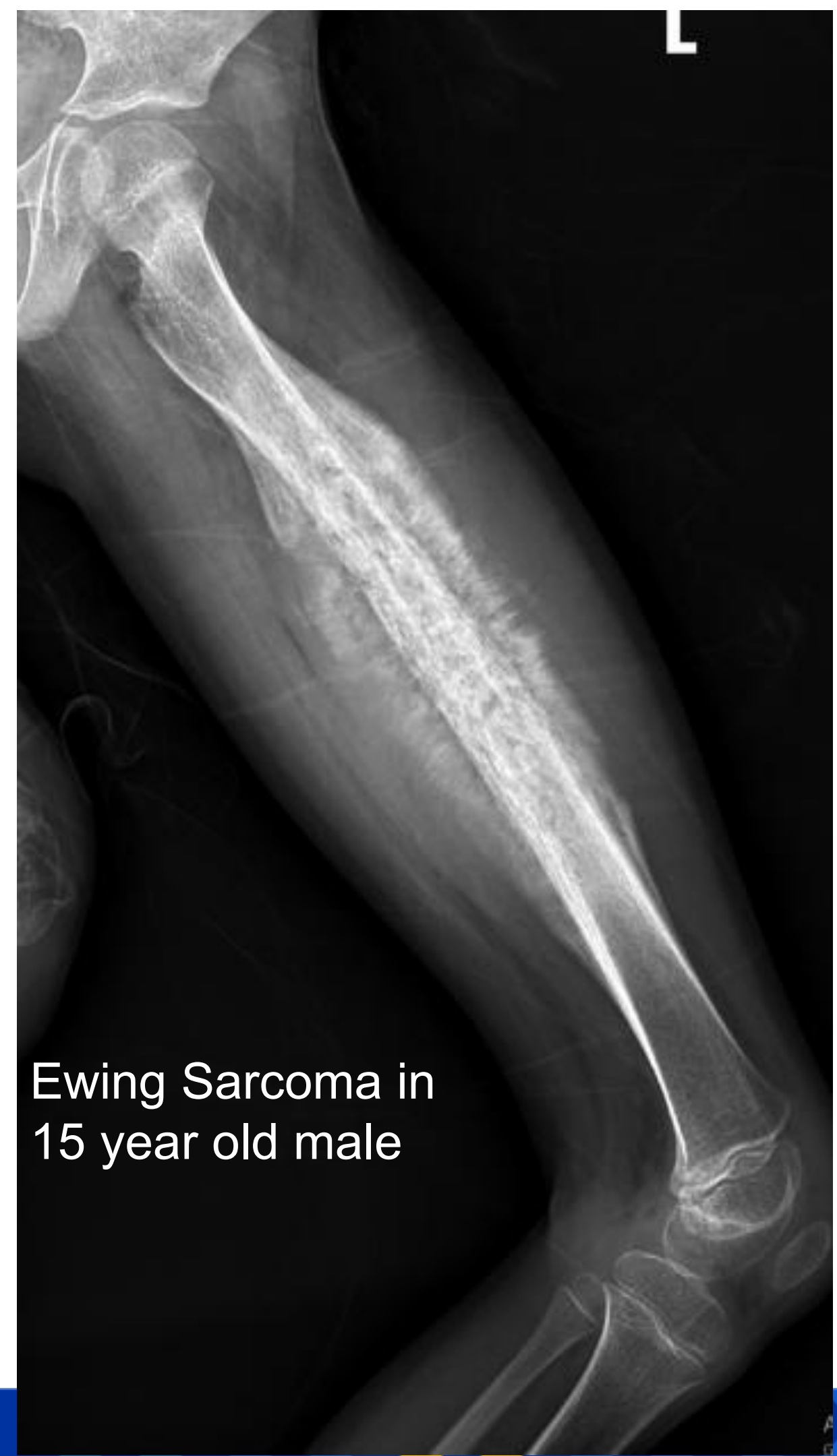
Often **multifocal**, unlike primary bone cancers.

Imaging findings:

- Lytic lesions (e.g., lung, kidney, thyroid metastases) → destructive, weak bone.
- Sclerotic lesions (e.g., prostate metastases).
- Mixed lesions (e.g., breast cancer).

Clinical Pearls

- Any fracture with minimal trauma in an adult >40 → suspect metastasis until proven otherwise.
- In a child or adolescent with suspicious bone lesion + fracture → think primary bone cancer.
- Always evaluate for systemic history (prior malignancy, systemic symptoms).
- MRI and biopsy are essential in diagnosis.



Ewing Sarcoma in
15 year old male

Name... That... Fracture!



Name... That... Fracture!

Transverse Fracture

Right tibial mid-diaphyseal transverse fracture, minimally displaced, without angulation or comminution.



Name... That... Fracture!



Name... That... Fracture!

Spiral Fracture

Left proximal-to-mid femoral diaphyseal spiral fracture, displaced medially with apex lateral angulation.



Name... That... Fracture!



Name... That... Fracture!

Comminuted Fracture

Proximal femoral diaphyseal comminuted fracture with significant displacement, apex lateral angulation, and multiple free fragments.



Name... That... Fracture!



Name... That... Fracture!

Segmental Fracture (fracture in which a bone is broken in two or more distinct places)

Segmental fracture of the femoral shaft with two distinct transverse fracture lines, resulting in a separated intermediate fragment. Distal fragment displaced medially with apex lateral angulation.



Name... That... Fracture!



Name... That... Fracture!

Intra-articular Fracture

Intra-articular distal radius fracture with dorsal and radial displacement of the distal fragment, mild dorsal angulation, and articular comminution.



Name... That... Fracture!



Name... That... Fracture!

Impacted Fracture

Impacted fracture of the femoral neck with minimal displacement and cortical overlap



Name... That... Fracture!



Name... That... Fracture!

Compression Fracture

Lumbar vertebral body compression fracture with loss of anterior height and wedge-shaped deformity



Name... That... Fracture!



Name... That... Fracture!

Stress Fracture (small, incomplete fracture caused by repetitive submaximal stress on bone, exceeding its ability to remodel)

Distal tibial stress fracture with periosteal reaction and transverse fracture line, non-displaced



Name... That... Fracture!



Name... That... Fracture!

Pathologic Fracture (Osteosarcoma)

Pathologic fracture of the proximal humerus through an osteosarcoma, with mixed lytic/sclerotic lesion, periosteal reaction, cortical destruction, and associated soft tissue mass.



Name... That... Fracture!



Name... That... Fracture!

Pathologic Fracture (Lytic Lesion)

Pathologic fracture through a lytic lesion in the mid-ulnar diaphysis, with cortical destruction and minimal displacement



Name... That... Fracture!



Name... That... Fracture!

Salter Harris Type IV Fracture

Salter-Harris Type IV fracture of the distal tibia with intra-articular extension and minimal displacement



Name... That... Fracture! (Pediatrics)



Name... That... Fracture! (Pediatrics)

Buckle (Torus) Fracture

Distal radial and ulnar metaphyseal buckle (torus) fractures, extra-articular and non-displaced



Name... That... Fracture! (Pediatrics)



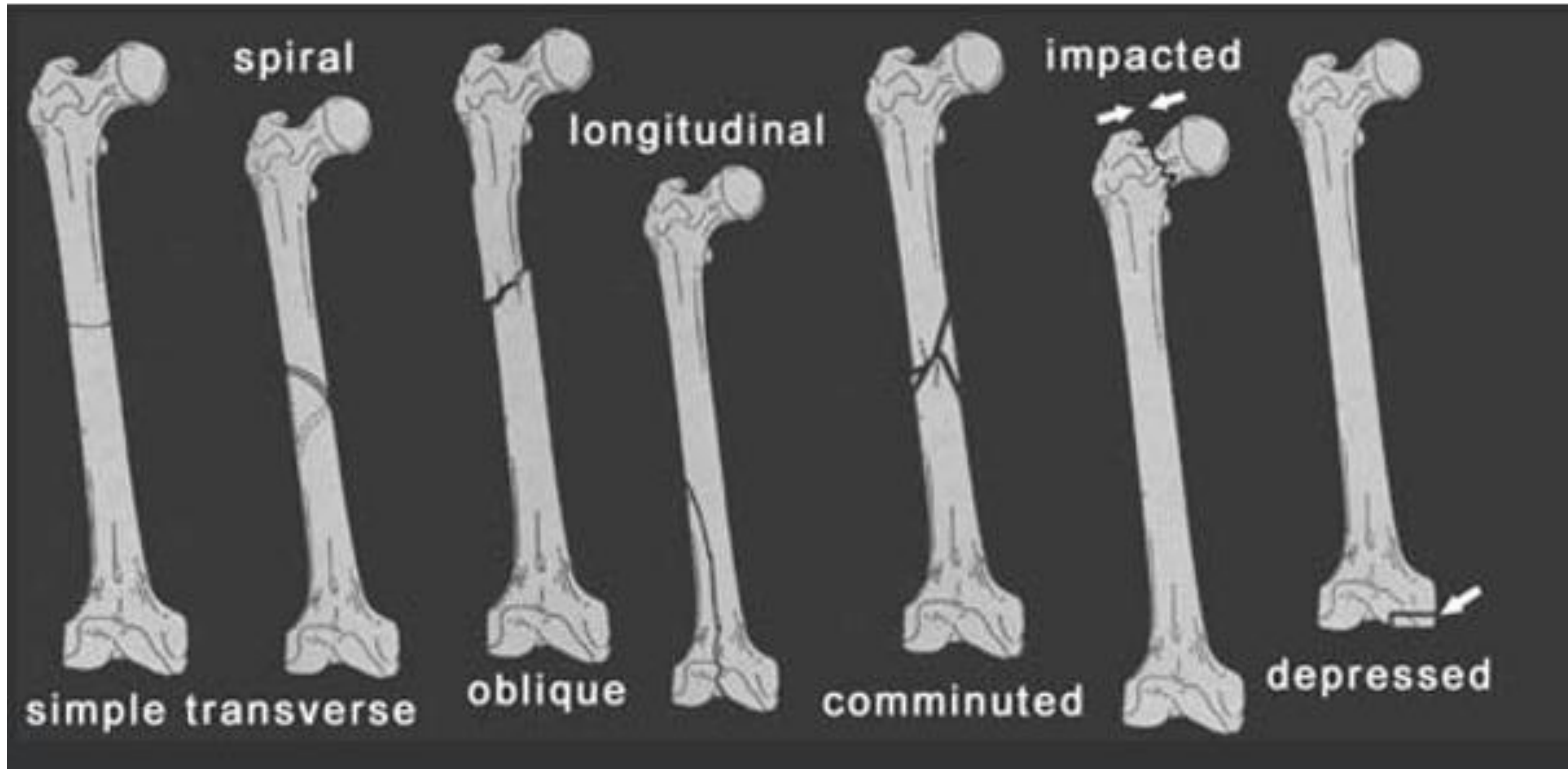
Name... That... Fracture! (Pediatrics)

Greenstick Fracture

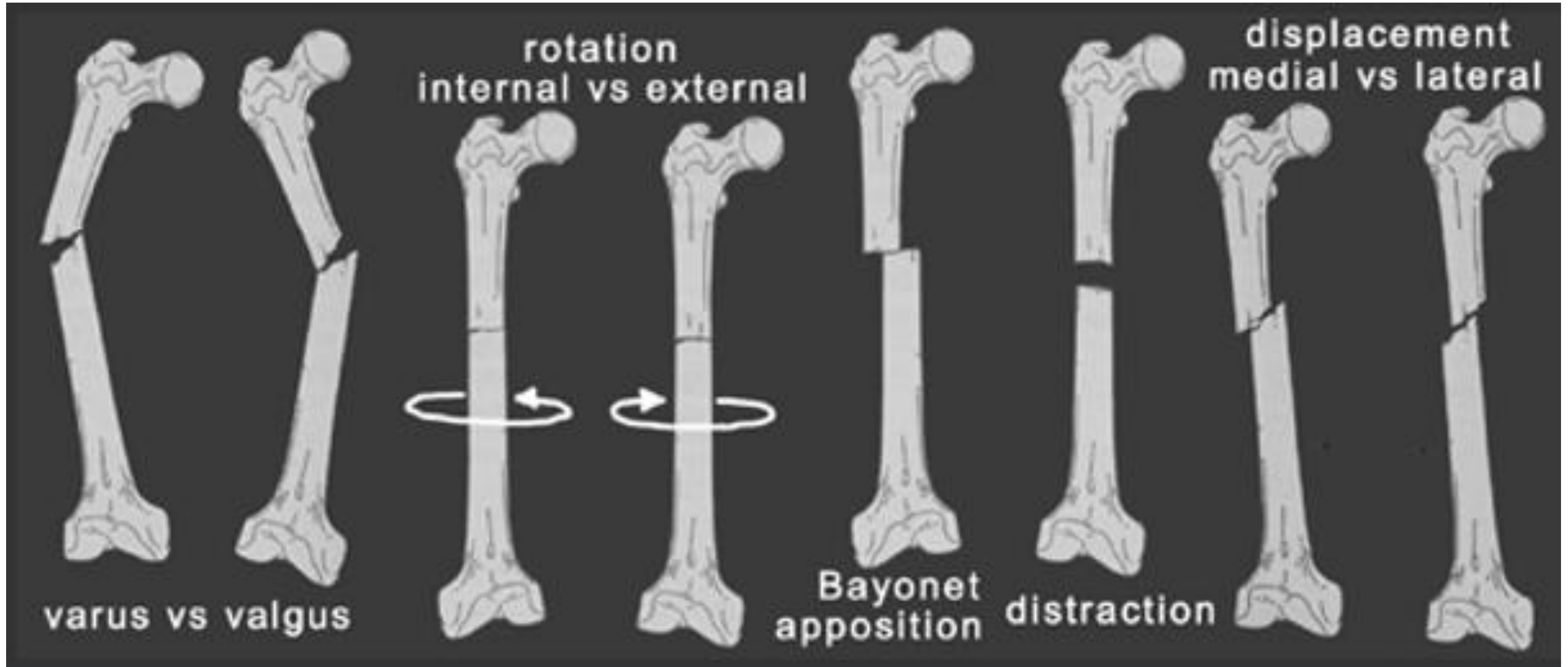
Distal radius and ulna greenstick fractures: incomplete fractures with cortical break on the tension side and intact cortex on the compression side, extra-articular, with mild angulation.



Fracture Reviews



Fracture Reviews



Fracture Reviews



Segmental

A type of comminuted fracture in which a completely separate segment of bone is bordered by fracture lines



Spiral

A multiplanar and complex fracture line



Intra-articular

The fracture line crosses the articular cartilage and enters the joint

Fracture Reviews



Impaction

A fracture that occurs when one bone hits or "impacts" an adjacent bone



Compression

A type of impaction fracture that occurs in the vertebrae, resulting in depression of the end plates



Depression

A type of impaction fracture that occurs in the knee when the femoral condyle strikes the softer tibial plateau

Fracture Reviews



Stress (fatigue)

A fracture in normal bone that has been subjected to repeated or cyclical loads that in and of themselves are not sufficient to cause a fracture



Stress (insufficiency)

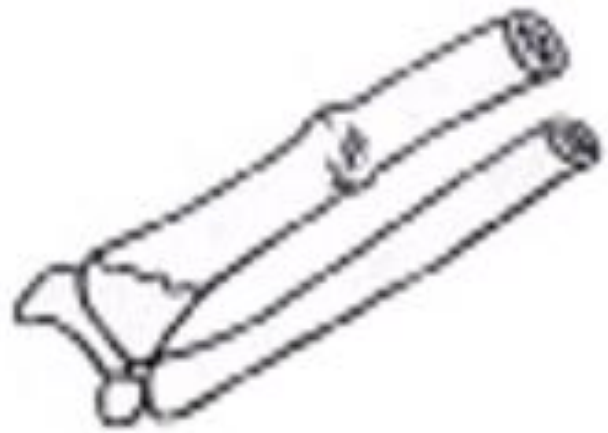
A fracture in weakened bone that has been subjected to a load insufficient to fracture normal bone



Pathologic

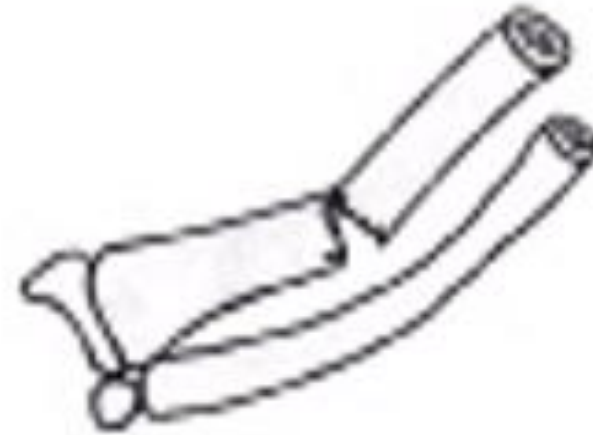
A fracture through bone weakened by tumor, metabolic bone disease, or osteoporosis

Fracture Reviews



Torus

An incomplete buckle fracture of one cortex, often seen in children



Greenstick

An incomplete fracture with angular deformity, seen in children

How Bones Heal

Bone is unique among tissues because it heals **by regeneration** (restoring normal structure and strength) rather than by scarring.

Healing can occur by two main mechanisms:

- **Primary (direct) bone healing** – when bone ends are anatomically aligned and stabilized.
- **Secondary (indirect) bone healing** – occurs with some motion at the fracture site (most common in natural healing).



Phases of Bone Healing

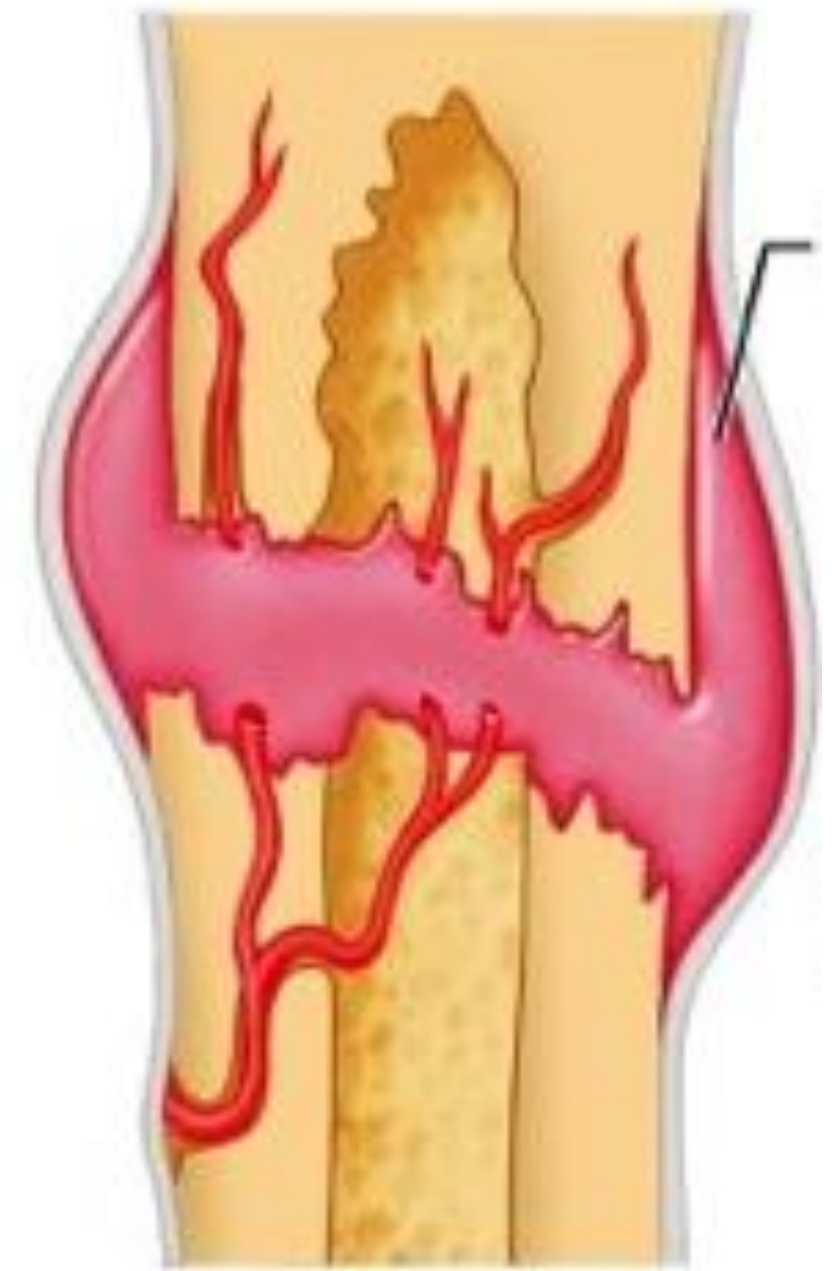
1. Inflammatory Phase (0–7 days)

Immediately after fracture:

- Blood vessels rupture → **hematoma** formation at the fracture site.
- Platelets release growth factors (PDGF, TGF- β , VEGF) that recruit inflammatory cells.
- **Neutrophils and macrophages** remove necrotic debris.

Biologic role: sets the foundation for repair by initiating angiogenesis and recruiting osteoprogenitor cells.

Radiologic appearance: fracture line clearly visible; soft tissue swelling.



① **Hematoma formation**

Phases of Bone Healing

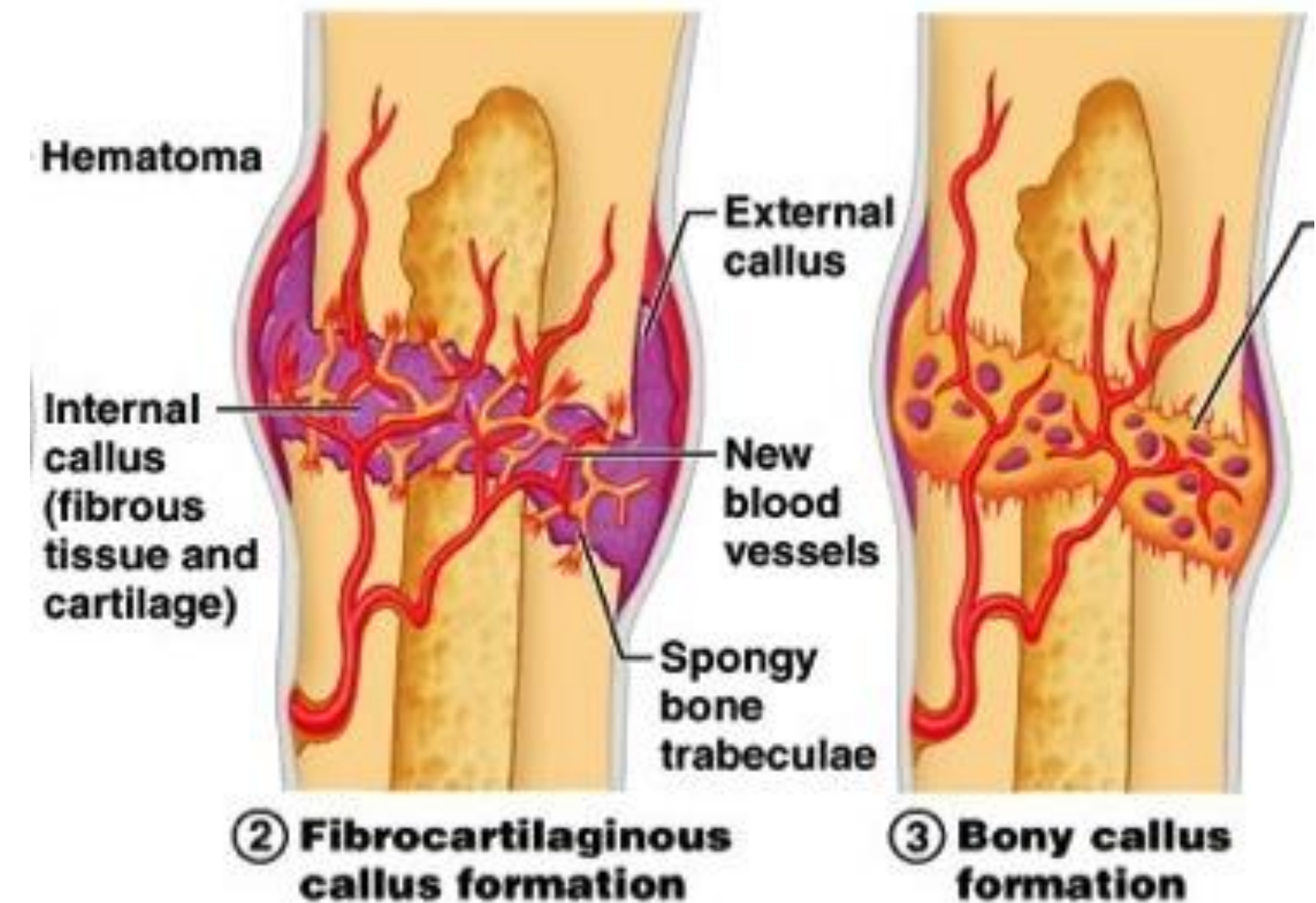
2. Reparative Phase (1–4 weeks)

Soft (fibrocartilaginous) callus formation:

- Fibroblasts and chondroblasts infiltrate the hematoma → form granulation tissue and fibrocartilage bridge.
- Provides temporary stability.

Hard (bony) callus formation:

- Osteoblasts lay down **woven bone** on the cartilage scaffold.
- Callus mineralizes and becomes radiographically visible (2–3 weeks post-injury).



Clinical correlation:

Pain and tenderness decrease

Motion improves

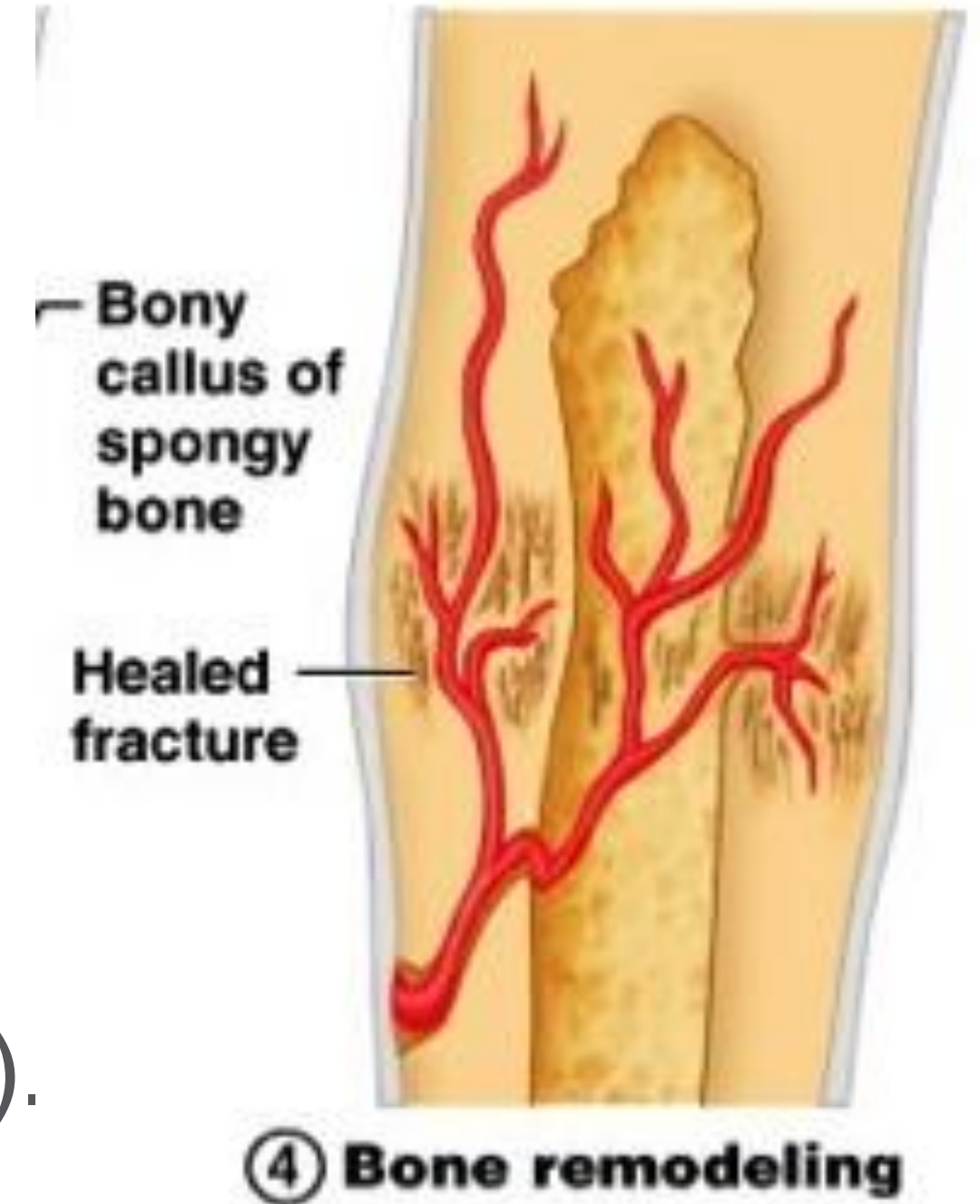
Phases of Bone Healing

3. Remodeling Phase (months to years)

Woven bone → **lamellar bone** through osteoclastic resorption and osteoblastic deposition.

- The medullary canal is reconstituted.
- Bone adapts to mechanical stress (Wolff's Law).

Radiologic appearance: gradual smoothing of callus and restoration of normal cortical contour.



Factors Affecting Bone Healing

Enhancing Factors

- Adequate blood supply
- Stable fracture fixation
- Good alignment
- Healthy periosteum
- Younger age

Delaying Factors

- Poor blood flow (e.g., open fractures, high-energy injury)
- Infection
- Smoking, diabetes, malnutrition
- NSAIDs or corticosteroids (inhibit prostaglandin-mediated healing)
- Bone loss or interposition of soft tissue

Clinical Case

COMLEX AND CLINICAL ROTATION HACK

A 10-year-old male is brought to the emergency department following a fall off his bike. He has difficulty moving his shoulder. Plain film radiographs and an angiogram are obtained and reveal a fracture of the surgical neck of his humerus and bleeding from the fracture site. From these results, it is apparent that this accident most likely caused damage to the:

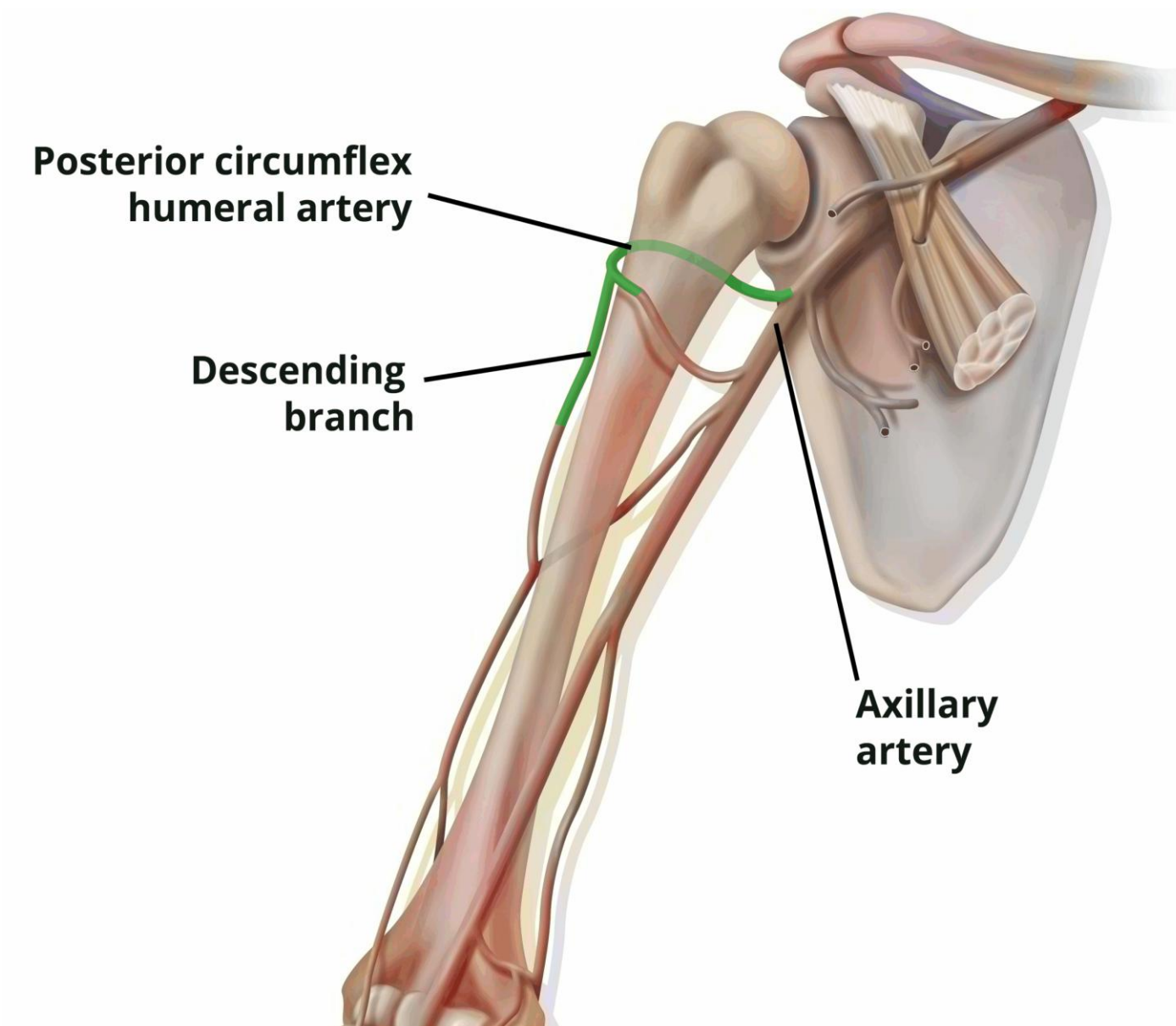
- A. axillary artery
- B. deep brachial artery
- C. posterior humeral circumflex artery
- D. radial artery
- E. subscapular artery

Clinical Case

COMLEX AND CLINICAL ROTATION HACK

C. posterior humeral circumflex artery

Fractures of the proximal humerus typically occur from a fall on an outstretched hand or a direct blow. The posterior circumflex humeral artery passes around the surgical neck of the humerus with the axillary nerve. Axillary nerve deficits manifest as decreased sensation over the deltoid area of the shoulder or loss of shoulder abduction. Knowing where the injury is relative to the bone or overlying structure is key to understanding which artery is injured.



Clinical Case

COMLEX AND CLINICAL ROTATION HACK

A 21-year-old male presents to the emergency department following an injury to his left arm during a college wrestling competition. He heard a "popping sound" following the takedown by his opponent. He has severe discomfort in his left upper extremity and will not allow his arm to be touched during physical examination. Vital signs and initial laboratory tests are within normal limits. A left upper extremity radiograph is obtained, as shown in the exhibit. The most likely motor deficit observed on physical examination would be

- A. extension of the fingers at the MCP joint
- B. flexion at the wrist
- C. flexion of the arm at the elbow
- D. flexion of the medial fingers
- E. opposition of the thumb



Clinical Case

COMLEX AND CLINICAL ROTATION HACK

A. extension of the fingers at the MCP joint

The peripheral nerve most commonly injured with a mid-diaphyseal fracture of the humerus is the radial nerve. This can lead to impaired function of the brachioradialis, supinator, triceps, and wrist/finger extensor muscles.



Clinical Pearls and Take Home Points

- ❖ **Describe fractures systematically:** include site, displacement, angulation, comminution, and joint involvement for clear, consistent communication.
- ❖ **Pediatric bone differs from adult bone:** the physis is the weak link; thick periosteum and strong remodeling lead to faster healing and unique fracture types.
- ❖ **Not all fractures are traumatic:** minimal-trauma fractures, especially in adults >40, often indicate underlying pathology such as metastasis.
- ❖ **Bone heals by regeneration:** inflammation → repair → remodeling; alignment and vascularity are key for recovery.
- ❖ **Imaging guides management:** match modality (X-ray, CT, MRI, US) and anatomy to mechanism for accurate interpretation and treatment planning.